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Project Proposal: Stock Price Prediction using Regression Models

The goal of this project is to apply statistical modeling techniques to forecast stock price behavior based on historical data. Specifically, I aim to implement and evaluate two predictive models: **linear regression** to estimate the next day's closing price, and **logistic regression** to classify whether the stock will move up or down the following day. These models will help assess whether past return patterns can provide predictive power for short-term stock price movements—an objective well-suited to regression-based statistical tools.

2. What key outcome variable(s) are being used to address this goal?

- For **Linear Regression**, the outcome variable will be the **next day's closing price**, treated as a continuous variable.
- For **Logistic Regression**, the outcome variable will be a **binary indicator** (1 = price goes up, 0 = price goes down) based on whether the next day's closing price increases relative to the current day's close.

3. What are some potential features or predictors that could be useful in the model?

I will engineer a set of features from time-series data that may have predictive value, including:

- **Daily return percentages** for the prior 5 days (lagged returns)
- **5-day moving average of returns**
- **5-day return volatility (standard deviation)**
- **Relative Strength Index (RSI)** over the past 5 days
- **Momentum indicator** (e.g., 5-day moving average minus 10-day moving average)
- **Previous day's trading volume**
- **Price range** (difference between high and low) for recent days

These features are selected based on common financial indicators and technical analysis principles, and they will be used to train and validate both models.

Dataset Information:

I will collect historical daily stock price data using the **Yahoo Finance API** (via libraries like yfinance in Python). The dataset will include OHLC (open, high, low, close) prices and volume for a selection of actively traded stocks, such as components of the S&P 500, over a multi-year horizon. I will derive the previous 5 days return from the dataset. Also, the Label “Up”, or “Down” will also be derived. This will ensure the models are trained on a sufficiently large and diverse dataset to evaluate performance.